

sets whose contributors are African businesses and organisation. Zindi is a great platform!

If you want to train the model on your laptop, then download the data else we can train it on the kernel itself. In this approach, we are going to follow the latter approach.

Preparing the kernel:

To prepare the kernel for the task use the blue new kernel button on the top right corner, then pick notebook from the Select Kernel Type. This enables us to do some interactive programming on the Jupyter Notebook. This creates a new private kernel and automatically adds the dogs' vs cats' dataset to your file path that is in the cloud. Do not forget to use your super processing power of the server the GPU for faster compute time. Now our kernel is ready for the image recognition process.

Importing Libraries and Image Dataset:

Now we need to import libraries such as numpy, matplotlib, Keras, scikit – learn, pandas, os and random. All these can be imported by using the import function of python. The image dataset is stored into three Zip files in our kernel:

1. Sample submission.csv:

This is a comma separated value (CSV) file, which is used for making a submission after training and testing your model on the test file.

2. test.zip:

This file contains the images that we're going to test our models on after training. It will let us know if our model has correctly learnt to differentiate dogs from cats.

3. train.zip:

This is the training dataset for our model that will be used to differentiate the dogs from the cats correctly.

Now we are going to use the os package to create a file path to our training and testing data.

1. First, we create two variables `train_dogs` and `train_cats` — the first one for all dog images and the other one for cat images. We write a python list comprehension which uses the command `os.listdir()` from the os library